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Restoring Vocal Fold Movement after Transection and Immediate Suturing of the Recurrent Laryngeal Nerve with Local Application of Basic Fibroblast Growth Factor: An Experimental Study in the Rat

Kazumi Motoyoshi, MD; Masamitsu Hyodo, MD; Takahiko Yamagata, MD; Kiyofumi Gyo, MD

Objective: **To evaluate the effects of basic fibroblast growth factor (bFGF) on the recovery of vocal fold movement and the attenuation of laryngeal muscle atrophy after transection of the recurrent laryngeal nerve (RLN).** ***Study Design:*** **Quantitative assessment of vocal fold movement using the video cassette recorder (VCR) image-analysis method and histologic examination of the laryngeal muscle.** ***Methods:*** **Fifty-eight Wistar rats underwent RLN transection and one of the following three procedures: 1) transection of the RLN alone (transection group, n = 18), 2) suture of the nerve stumps followed by local administration of phosphate-buffered saline (PBS) solution using an osmotic pump (PBS group, n = 20), or 3) suture of the nerve stumps followed by local administration of bFGF (FGF group, n = 20). Vocal fold movements were recorded with VCR by way of a rigid endoscope, and the VCR images were analyzed on a computer. Histologic changes in the thyroarytenoid (TA) muscle were evaluated by measuring the cross-sectional area of the muscle and average size of muscle fibers.** ***Results:*** **In the transection group, vocal fold movement did not recover, and atrophy of the TA muscle gradually progressed after sectioning the nerve. In contrast, vocal fold movement as assessed by VCR image-analysis recovered in some cases in the immediate suturing groups, more markedly in the FGF group (34.1 ± 29.1%) than in the PBS group (5.5 ± 7.9%) ($P < .05$). Histologically, atrophy of the laryngeal muscle was significantly attenuated by the local administration of bFGF.** ***Conclusion:*** **bFGF facilitates regeneration of the transected RLN and attenuation of intrinsic laryngeal muscle atrophy, thereby restoring laryngeal function.** ***Key Words:*** **Basic fibroblast growth factor, recurrent laryngeal nerve, laryngeal muscle, vocal fold movement, osmotic pump.**

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INTRODUCTION

Transection of the recurrent laryngeal nerve (RLN), which sometimes occurs as a consequence of a neck injury or thyroid surgery, causes acute severe dysphonia and aspiration caused by RLN paralysis. The advocated treatment is immediate suturing of the nerve stumps or grafting the nerve under an operating microscope. This does not always result in good functional recovery because regeneration of the nerve fiber at the site of injury is often incomplete, and atrophy of the laryngeal muscle progresses until there is complete recovery of innervation. Furthermore, misdirection of the regenerating nerve fiber may induce paradoxical movement of the vocal folds.

Recent experimental studies indicate that neurotrophic factors, such as nerve growth factor (NGF), ciliary neurotrophic factor (CNTF), fibroblast growth factor (FGF), and glial cell line-derived neurotrophic factor (GDNF), promote nerve regeneration, including regeneration of the RLN. In 1999, Komori compared the potential of these neurotrophic factors with regenerate the RLN in rats and concluded that basic FGF (bFGF) best facilitated axonal regeneration of the injured RLN.[1] bFGF, which belongs to the FGF family, is a multipotential heparin-binding protein with marked action on central and peripheral neurons during regeneration.[2] It is especially effective in preventing neuronal death[3,4] and promoting neuronal sprouting[5,6] after transection of the nerve when administered locally to the site of an injured nerve or the nucleus.

This study was performed to examine whether immediate nerve suturing and the local administration of bFGF

From the Department of Otolaryngology, Ehime University School of Medicine, Shigenobu-cho, Onsen-gun, Ehime, Japan.

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Send Correspondence to Dr. Kazumi Motoyoshi, Department of Otolaryngology, Ehime University School of Medicine, Shigenobu-cho, Onsen-gun, Ehime, 791–0295, Japan.

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after transection of the RLN facilitates the recovery of vocal fold movement by promoting regeneration of the damaged nerve and preventing atrophy of the intrinsic laryngeal muscle in an animal model.

MATERIALS AND METHODS

Eleven-week-old male Wistar rats were used in this study. All the animals were housed at a constant temperature (22°C) with a 12-hour light-dark cycle and given food and water as desired. All experiments were conducted in accordance with the guidelines of the Animal Care and Use Committee in compliance with the Animal Research Center of Ehime University School of Medicine, Ehime, Japan.

Animal Model

The animals anesthetized with an intramuscular injection of ketamine hydrochloride (45 mg/kg), and the surgical procedures were performed under aseptic conditions. After subcutaneous injection of 1% lidocaine hydrochloride into the anterior neck, a vertical skin incision was made, and the left RLN was carefully exposed under an operating microscope (Olympus OME, Tokyo, Japan). The RLN was cut at the level of the eighth tracheal ring with microscissors. The animals were then subjected to one of the following three procedures: 1) nerve transection alone (transection group, n = 18), 2) nerve transection and local administration of phosphate-buffered saline solution (PBS) (PBS group, n = 20), or 3) nerve transection and local administration of bFGF (FGF group, n = 20). In the transection group, the distal cut-end of the nerve was ligatured with 8–0 nylon, and the proximal end was embedded in the sternohyoid muscle to prevent reinnervation. In the PBS and FGF groups, the transected nerve stumps were immediately approximated and sutured with 10–0 nylon.

In the PBS group, the animals were given an infusion by way of an implanted osmotic pump (Alzet osmotic pump, model 2002, Alza Corporation, Palo Alto, CA) filled with 0.1 mol/L PBS. In the FGF group, the osmotic pump was filled with a 200 μL solution containing 1 μg human recombinant bFGF (Sigma, Saint Louis, MO) and 0.1% bovine serum albumin (BSA) in PBS. A long silicone tube (0.5 mm in inner diameter, 1.0 mm in outer diameter, and 30 mm in length) was connected to the pump at one end, and the other end was plugged with a glass bead. A short silicone tube (1.0 mm in inner diameter, 2.0 mm in outer diameter, and 3.0 mm in length) was attached along the long tube at its distal end. After splitting the short tube longitudinally, we created a hole 0.5 mm in diameter was to establish a delivery channel for PBS or bFGF from the pump to the lumen of the short tube. The short tube was placed around the RLN at the suture site to deliver solution from the pump. The pump was secured in a subcutaneous pocket created with blunt dissection in the right anterior neck (Fig. 1). To activate the bFGF, the pump was incubated at 37°C for 4 hours before implantation.

The solution was released at a rate of 0.5 μL/hour at the nerve suture site for 14 days. On day 15, the pump was removed under general anesthesia. The residual volume of solution was measured to determine the volume delivered during implantation. After completion of these procedures, the skin incision was closed, and the animal was allowed to recover in an approved animal care facility until subsequent study 30, 60, or 90 day after transection on the nerve. In these later studies, the examiners were blinded to the contents of implanted pumps.

Evaluation of Vocal Fold Movement

Vocal fold movement was evaluated in the animals in the PBS (n = 8 of 20) and FGF (n = 8 of 20) groups 90 days after transection of the nerve. The anesthetized rat was secured on an operating platform in the supine position. A rigid endoscope

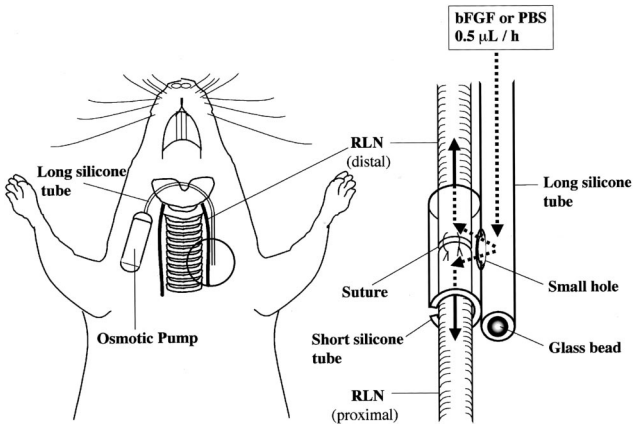


Fig. 1. Schematic drawing of the osmotic pump that was used for local application of phosphate-buffered saline (PBS) or basic fibroblast growth factor (bFGF). Flow route of test solution (arrow). RLN = recurrent laryngeal nerve.

(SFD, Nagashima Medical Instruments Co., Tokyo, Japan) was inserted orally and fixed in position to provide the best view of the larynx. Vocal fold movements during spontaneous respiration were recorded with a video cassette recorder (VCR, AU-W35R, Panasonic, Osaka, Japan) by way of a charge-coupled device camera (WV-KS152, Panasonic, Osaka, Japan) coupled to the endoscope.

Vocal fold movements were evaluated by analyzing the VCR images. As shown in Figure 2, A and B, two frames of the vocal fold picture during spontaneous respiration were selected and captured with a computer (Power Macintosh G4, Apple Computer, Cupertino, CA), one in its fully adducted position and the other in its fully abducted position. The free edges of the vocal fold in each frame were outlined using image-capturing software (Photoshop, Adobe, San Jose, CA) (Fig. 2, C and D). Traces of the vocal fold movement were obtained by calculating the size of the quadrilateral encompassed by the free edge of the vocal fold in the abducted and adducted positions and the lines connecting the anterior and posterior ends of the vocal fold in these two positions using a National Institutes of Health (NIH) Image (Fig. 2E). The laryngeal function was evaluated by com-

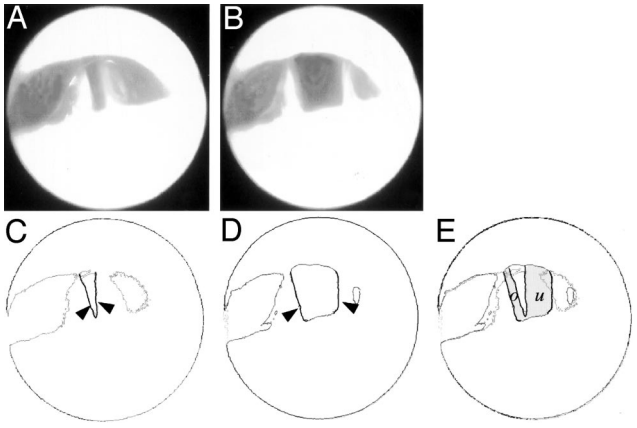


Fig. 2. Images and traced outline of the vocal fold at full adduction (**A**, **C**) and full abduction (**B, D**). Free edges of the vocal fold (arrowheads). (**E**) Composite of images c and d. Traces of vocal fold movement on the operated (O) and unoperated (U) sides.

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paring the area of the vocal fold movement on the operated (O) and unoperated (U) sides (O /U ratio).

Histologic Study

After euthanasia with a lethal dose of sodium pentobarbital on day 30, 60, or 90, the larynx was harvested and immediately frozen with liquid nitrogen. The tissue was sectioned sequentially in the coronal plane at a thickness of 15 μm using a cryostat at −20°C. The sections were processed for hematoxylin-eosin staining and observed under a light microscope (Optiphot, Nikon Inc., Tokyo, Japan) equipped with a digital still camera (HC-300, FujiFilm, Tokyo, Japan). Microscopic images of the thyroarytenoid (TA) muscle at the level of the vocal process were imported into a computer (Macintosh 7600, Apple Computer). The cross-sectional area of the TA muscle and its average muscle fiber size were measured by tracing outlines of the microscopic images using a NIH Image. More than 240 fibers were measured to determine the average size in each animal. Histologic changes in the TA muscle were evaluated by comparing the results between the operated (O) and unoperated (U) sides (O /U ratio). Six normal animals served as controls in the histologic study.

Statistical Analysis

The unpaired Student t test was used to compare vocal fold movement, the cross-sectional area of the TA muscle, and average size of muscle fibers between the FGF and PBS groups. Statistical analyses were performed using StatView (Abacus Concepts Inc., Berkeley, CA). A P value less than .05 was taken to indicate significance.

RESULTS

Vocal Fold Movement

Videoendoscopic observation of the larynx before RLN transection showed symmetrical vocal fold movement during respiration in all animals. They tolerated implantation of the drug-delivery system for 90 days without infection, exposure of the osmotic pump, or detachment of the silicone tube.

Vocal fold movement was analyzed in eight animals, respectively, in the PBS and FGF groups. Recovery of vocal fold movement was seen in seven of eight (87.5%) animals in the FGF group and three of eight (37.5%) animals in the PBS group 90 days after the operation, although the amplitude of the recovered vocal fold movement varied. Figure 3 summarizes the O/U ratio during respiration. The ratio was 34.1 ± 29.1% in the FGF group and 5.5 ± 7.9% in the PBS group; the difference was statistically significant ($P < .05$), indicating that recovery of the vocal fold movement was better in the FGF group.

Histologic Study

Representative cross-sectional findings of the TA muscle are shown in Figure 4. In the normal control, there was no substantial difference by side (Fig. 4A), and the mean cross-sectional area of the muscle was 1.44 ± 0.25 mm^2 (n = 6). In contrast, the TA muscle showed marked atrophy in the transection group (Fig. 4B). In the PBS and FGF groups, mild deformity of the muscle outline, casued by atrophy of muscle fibers, was observed on the operated side (Fig. 4, C and D). As shown in Figure 5, the O/U ratio of the muscle area decreased to approximately 90% on day 30 in all groups. On day 90, the decrease was 74.0 ± 3.3%

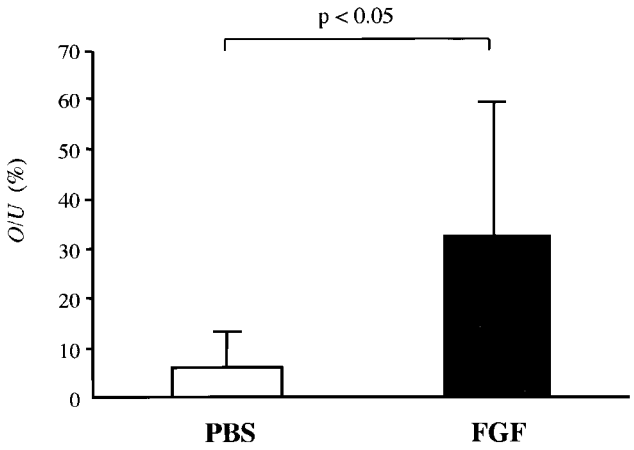


Fig. 3. Comparison of vocal fold movement 90 days after the procedure between the fibroblast growth factor (FGF) and phosphate-buffered saline (PBS) groups. Vocal fold movement was significantly larger in the FGF group (n = 8) than in the PBS group (n = 8).

in the transection group (n = 6). In contrast, the ratio recovered gradually to 98.0 ± 4.1% in the FGF group (n = 8), whereas it was stable (90.2 ± 2.1%) in the PBS group (n = 8). The O/U ratio in the FGF group was significantly larger than those in the transection ($P < .01$) and PBS groups ($P < .05$) on day 90.

Figure 6 shows a cross-section of the TA muscle. On the unoperated side, there were no substantial differences in the appearance of muscle fibers in the different groups (Fig. 6, A, B, and C). In contrast, the muscle was atrophic on the operated side (Fig. 6, D, E, and F), especially in the transection group. Sequential changes in the ratio of muscle fiber size on the operated and unoperated sides (O/U

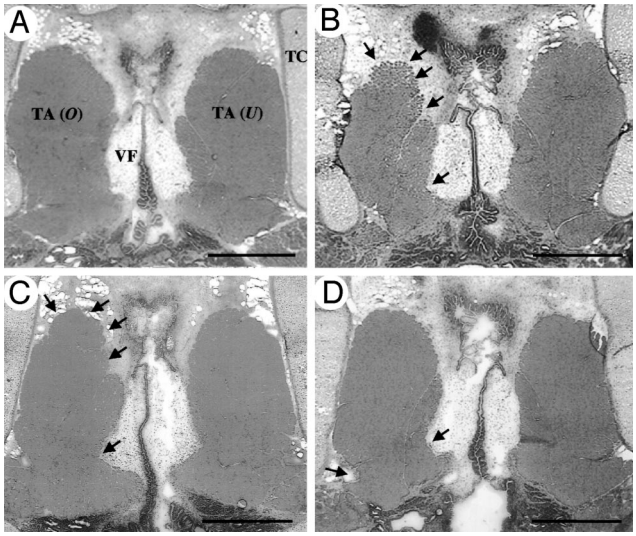


Fig. 4. The microscopic findings in the thyroarytenoid muscle (TA) (hematoxylin-eosin staining, magnification ×40). (**A**) Normal control. (**B**) Transection group. (**C**) Phosphate-buffered saline (PBS) group. (**D**) Basic fibroblast growth factor (bFGF) group. Atrophy of the TA on the operated side (arrows). TC = thyroid cartilage; VF = vocal fold; O = operated side; U = unoperated side. Scale bars = 1 mm.

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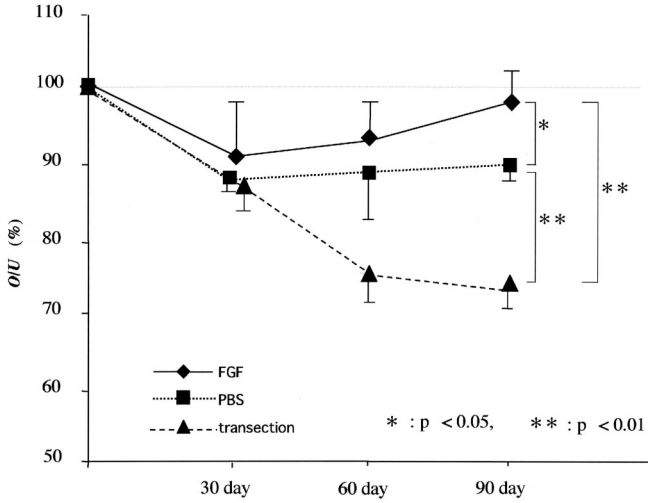


Fig. 5. Sequential change in operated/unoperated (O/U) ratio for the entire muscle area of the thyroarytenoid muscle (TA) muscle. Horizontal scale indicates the number of days after transection and repair of the left recurrent laryngeal nerve. n = 6 for each symbol except 90-day phosphate-buffered saline (PBS) and basic fibroblast growth factor (bFGF) (n = 8).

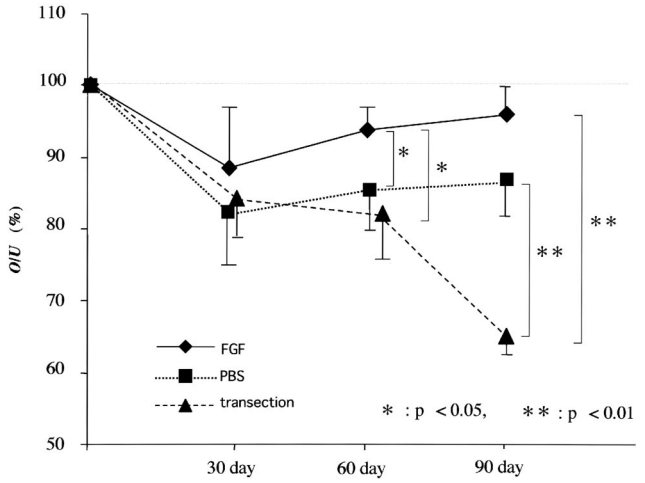


Fig. 7. Sequential changes in the operated/unoperated (O/U) ratio of the average size of muscle fibers. n = 6 for each symbol expect 90-day phosphate-buffered saline (PBS) and basic fibroblast growth factor (bFGF) (n = 8).

ratio) are shown in Figure 7. The average size of the muscle fibers before the operation was 626.9 ± 174.0 μm^2 (n = 6), and this decreased gradually with time after sectioning the RLN. On day 90, the O/U ratio of the average size was 65.2 ± 2.5, 86.8 ± 4.8, and 95.9 ± 2.9%, respectively, in the transection (n = 6), PBS (n = 8), and FGF (n = 8) groups. The differences were statistically significant. The number of TA muscle fibers was also compared between the operated and unoperated sides. The average number in an intact TA muscle was 1675 ± 25 (n = 6). On day 90, the O/U ratio of fiber number was 94.3 ± 1.7, 96.2 ± 2.8, and 99.5 ± 2.0%, respectively, in the transection (n = 6), PBS (n = 8), and FGF (n = 8) groups. The differences were not statistically significant.

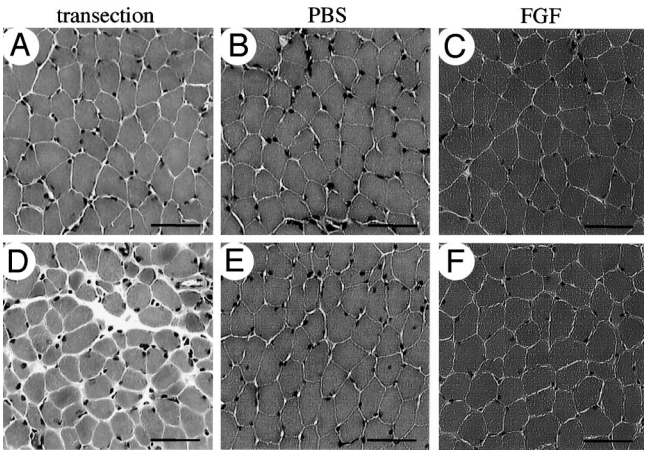


Fig. 6. Sections of thyroarytenoid muscle (TA) muscles 90 days after the procedure (hematoxylin-eosin staining). Upper row (**A, B, C**) indicates TA on the unoperated side, and lower row (**D, E, F**) indicates TA on the operated side. Scale bars = 50 μm.

DISCUSSION

Clinically, complete recovery of vocal fold movement after transection of the RLN is often difficult to obtain, despite intensive treatment. Once the nerve is severed, not all the nerve fibers regenerate perfectly, even when the nerve stumps are sutured immediately. In an experimental study on recovery of the cut and sutured RLN in the dog, Shin[7] reported that evoked myogenic potentials from the laryngeal muscles were low in amplitude, although the latency and wave form were within the normal range. He attributed these findings to incomplete recovery of the number of regenerated nerve fibers. Mu and Yang[8] reported similar findings in experiments with dogs. Misdirection of the regenerating nerve fiber is thought to be at least partly responsible for the incomplete recovery of vocal fold movement.[9–12] Because the RLN innervates both the abductor and adductor muscles of the larynx, regenerating RLN fibers may erroneously extend and innervate functionally different muscles after transection, resulting in paradoxical movement of the vocal fold. Furthermore, atrophy of the laryngeal muscle, which inevitably occurs with long-term denervation, may be responsible for poor mobility of the vocal fold.

Many experimental studies have examined means by which to facilitate recovery of the transected RLN. Gordon and McCabe[13] approximated the RLN immediately after transection in the dog and observed the recovery of vocal fold movement exclusively for adductive motion; they attributed this to the prevalence of adductive fibers. Adductive fibers in the RLN outnumber abductive fibers by 4:1 in the cat.[14] In related reports, the recovery of imperceptible adductive vocal fold movement was low, ranging from 0% to 29%.[15–18] In our study, vocal fold movement recovered in seven of eight (87.5%) animals in the bFGF group and three of eight (37.5%) animals in the PBS group after 90 days. Furthermore, we observed both adduction and abduction of the vocal fold in some animals treated with bFGF. These results suggest that bFGF facilitates regen-

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eration of the transected RLN, restoring laryngeal function, including the abductive function. Fujimoto et al.[19] investigated the effects of bFGF on nerve regeneration using electron microscopy and showed that a regenerating axon grew 1.5 to 2 times faster when the nerve stump was treated with bFGF versus PBS in the rat in vivo. According to Sunderland,[20] it takes 3 to 20 days for regenerating nerve fibers to pass through the sutured site; then, the fibers extend peripherally at the rate of 1 to 5 mm per day. As the distance between the lesion and the TA muscle is approximately 10 mm in our animal model, we estimate that the time necessary for a regenerating nerve fiber to reach the muscle is within 30 days. Therefore, a 90-day follow-up period was thought sufficient to observe the outcome of RLN palsy.

In this study, we administered bFGF locally using an osmotic pump, which enabled long-term application of the agent directly at the lesion.[21,22] The appropriate concentration to maximize nerve regeneration differs with animal species, administration route, application speed, etc.[1,5,19,23,24] In a preliminary experiment, 5 μg/mL bFGF diluted in PBS proved best for application in our animal model. Therefore, we set the concentration of bFGF as 5 μg/mL and supplied 60 ng/day for 14 days. The manufacturers have demonstrated that the biologic activity of bFGF remains unchanged at 37°C for up to 4 weeks.[25]

As shown in Figures 5 and 7, the local administration of bFGF significantly attenuated atrophy of the TA muscle. The TA muscle is an adductor muscle and is thought to be the intrinsic laryngeal muscle that undergoes the most severe degeneration after resection of the RLN.[26] Although the reduced atrophy of the TA muscle was mainly caused by the fast recovery of RLN innervation caused by the effect of bFGF, the bFGF may also have worked directly on the muscle. However, there is no evidence for this. Helgren et al.[27] administered CNTF, an analogue of bFGF, to denervated skeletal muscle and demonstrated that it reduced muscle atrophy markedly. Furthermore, immunocytochemical[28] and molecular biologic[29] studies have demonstrated the expression of endogenous bFGF in injured muscle undergoing regeneration. Lefaucheur et al.[30] injected bFGF-neutralizing antibody into skeletal muscle after denervation and showed that the number of regenerating muscle fibers decreased. These findings suggest that bFGF is a myotropic agent that acts directly on the laryngeal muscle. In our study, bFGF might have leaked from the feeding tube and then spread directly to the intrinsic laryngeal muscles. This hypothesis requires further investigation, including assessment of the direct action bFGF on denervated muscle fibers.

In conclusion, the local application of bFGF at the site of the lesion improves recovery of vocal fold movement after transection and immediate repair of the RLN. It is thought to act by both facilitating regeneration of the damaged nerve fibers and by preventing denervation-induced atrophy of the laryngeal muscle, thereby improving the contractile force of the intrinsic laryngeal muscles. Another possible explanation is that bFGF minimizes the misdirection on the nerve fibers. No side effects related to the local use of bFGF were noted. Further studies are necessary to examine the effects of bFGF, such as misdirected reinnervation and the long-term outcome of laryngeal function.

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BIBLIOGRAPHY

1. Komori M. Effects of neurotrophic factors on regeneration of denervated recurrent laryngeal nerve in the rat. *Pract Otol (Kyoto)* 1999;92:677–686.
2. Unsicker K, Grothe C, Ludecke G, et al. Fibroblast growth factors: their roles in the central and peripheral nervous system. In: Loughlin SE, Fallon JH, eds. *Neurotrophic Factors*. New York: Academic Press, 1993:313–338.
3. Anderson KJ, Dam D, Lee S, Cotman CW. Basic fibroblast growth factor prevents death of lesioned cholinergic neurons in vivo. *Nature* 1988;332:360–361.
4. Otto D, Unsicker K, Grothe C. Pharmacological effects of nerve growth factor and fibroblast growth factor applied to the transsectioned sciatic nerve on neuron death in adult rat dorsal root ganglia. *Neurosci Lett* 1987;83:156–160.
5. Danielsen N, Pettmann B, Vahlsing HL, et al. Fibroblast growth factor effect's on peripheral nerve generation in a silicone chamber model. *J Neurosci Res* 1988;20:320–330.
6. Gerney ME, Yamamoto H, Korn Y. Induction of motor neuron sprouting in vivo by ciliary neurotrophic factor and basic fibroblast growth factor. *J Neurosci* 1992;12:3241–3247.
7. Shin T. Regeneration of recurrent laryngeal nerve after injury. *Otologica (Fukuoka)* 1971;17:117–121.
8. Mu LC, Yang SL. Electromyographic study on end-to-end anastomosis of the recurrent laryngeal nerve in dogs. *Laryngoscope* 1990;100:1009–1017.
9. Nahm I, Shin T, Watanabe H, Maeyama T. Misdirection of injured recurrent laryngeal nerve in the cat. *Am J Otolaryngol* 1993;14:43–48.
10. Siribodhi C, Sundmäker W, Atkins JP, Bonner FJ. Electromyographic studies of laryngeal paralysis and regeneration of laryngeal motor nerves in dogs. *Laryngoscope* 1963; 73:148–164.
11. Hiroto I, Hirano M, Tomita H. Electromyographic investigations of human vocal paralysis. *Ann Otol Rhinol Laryngol* 1968;77:296–304.
12. Tashiro T. Experimental studies on the reinnervation of larynx after accurate neurorrhaphy. *Laryngoscope* 1972;82: 225–236.
13. Gordon JH, McCabe BF. The effect of accurate neurorrhaphy on reinnervation and return of laryngeal function. *Laryngoscope* 1968;78:236–250.
14. Gacek R, Malmgren LT, Lyon MJ. Localization of adductor and abductor motor nerve fibers to the larynx. *Ann Otol Rhinol Laryngol* 1977;86:770–776.
15. Boles R, Fritzell B. Injury and repair of the recurrent laryngeal nerve in dogs. *Laryngoscope* 1969;79:1405–1418.
16. Tomita H. An electromyographic study of recurrent laryngeal nerve paralysis. *J Otolaryngol Jpn* 1967;70:963–985.
17. Dedo HH. Electromyographic and visual evaluation of recurrent laryngeal nerve anastomosis in dogs. *Ann Otol Rhinol Laryngol* 1971;80:664–668.
18. Shiotani A, Flint PW. Modulation of myosin heavy chains in rat laryngeal muscles after denervation. *Laryngoscope* 2001;111:472–477.
19. Fujimoto E, Mizoguchi A, Hanada K, et al. Basic fibroblast growth factor promotes extension of regenerating axons of peripheral nerve: in vivo experiments using a Schwann cell basal lamina tube model. *J Neurocytol* 1997;26:511–528.
20. Sunderland SS. Regeneration of the axon and associated changes. In: Sunderland SS, ed. *Nerves and Nerve Injuries*. New York: Churchill Livingstone, 1978:108–132.
21. Newman JP, Verity AN, Hawatmeh S, et al. Ciliary neuro-

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trophic factor enhances peripheral nerve regeneration. *Arch Otolaryngol Head Neck Surg* 1996;122:399–403.
22. Lewin SL, Utley DS, Cheng ET, et al. Simultaneous treatment with BDNF and CNTF after peripheral nerve transection and repair enhances rate of functional recovery compared with BDNF treatment alone. *Laryngoscope* 1997;107:992–999.
23. Grothe C, Wewetzer K, Langrenge A, Unsicker K. Effect of basic fibroblast growth factor on survival and choline acetyltransferase development of spinal cord neurons. *Dev Brain Res* 1991;62:257–261.
24. Ide C, Tohyama K, Tajima K, et al. Long acellular nerve transplants for allogeneic grafting and the effects of basic fibroblast growth factor on the growth of regenerating axons in dogs. *Exp Neurol* 1998;154:99–112.
25. Edelman ER, Mathiowitz E, Langer R, Klagsbrum M. Controlled and modulated release of basic fibroblast growth factor. *Bio Mat* 1991;12:619–626.
26. Iizuka T. Experimental studies on the nerve interception and atrophy of the intrinsic muscles of the larynx. *J Otolaryngol Jpn* 1965;69:176–195.
27. Helgren ME, Squinto SP, Davis HL, et al. Trophic effect of ciliary neurotrophic factor on denervated skeletal muscle. *Cell* 1994;76:493–504.
28. Anderson JE, Liu L, Kardami E. Distinctive patterns of basic fibroblast growth factor (bFGF) distribution in degenerating and regenerating areas of dystrophic (mdx) striated muscles. *Dev Biol* 1991;147:96–107.
29. Guthridge M, Wilson M, Cowling J, et al. The role of basic fibroblast growth factor in skeletal muscle regeneration. *Growth Factors* 1992;6:53–63.
30. Lefaucheur JP, Së bille A. Muscle regeneration following injury can be modified in vivo by immune neutralization of basic fibroblast growth factor, transforming growth factor β1 or insulin-like growth factor I. *J Neuroimmunol* 1995; 57:85–91.

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